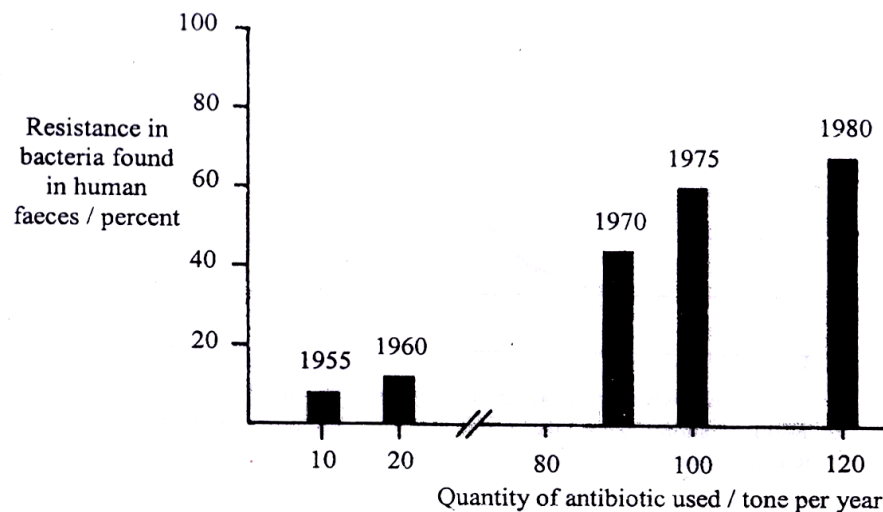


Question 1

Antibiotic medications are used to kill bacteria, which can cause illness and disease. They have made a major contribution to human health. Many diseases that once killed people can now be treated effectively with antibiotics. However, some bacteria have become resistant to commonly used antibiotics.

Antibiotic resistant bacteria are bacteria that are not controlled or killed by antibiotics. Due to mutation, they are able to survive and even multiply in the presence of an antibiotic. Most infection-causing bacteria can become resistant to at least some antibiotics. Bacteria that are resistant to many antibiotics are known as multi-resistant organisms (MROs).

The bar chart shows the levels of resistance of human faecal bacteria to antibiotics in relation to the amounts of antibiotics used in each of a number of years from 1955 to 1980.



a) Describe the relationship between the incidence of resistance and the use of antibiotics from 1955 to 1980. [2]

Incidence of resistance increases with increased use of antibiotics;

- **QV: Resistance found in bacteria found in human faeces increased from about 8% to 70% when quantity of antibiotics increased from 10 (in 1955) to 120 tone/yr (in 1980);;**

b) Explain how natural selection has led to the emergence of such antibiotic resistant bacteria. [4]

-Variation: different variation of bacteria some resistant to antibiotic some not

-Selection pressure: Antibiotic

-Selective advantage: Bacteria that is resistant to antibiotic

-Bacteria survived to produce offspring by binary fission and pass on their antibiotic resistance gene to the next generation.

[Total: 6 marks]

Question 2

- Figure 2.1 shows a haemoglobin molecule.

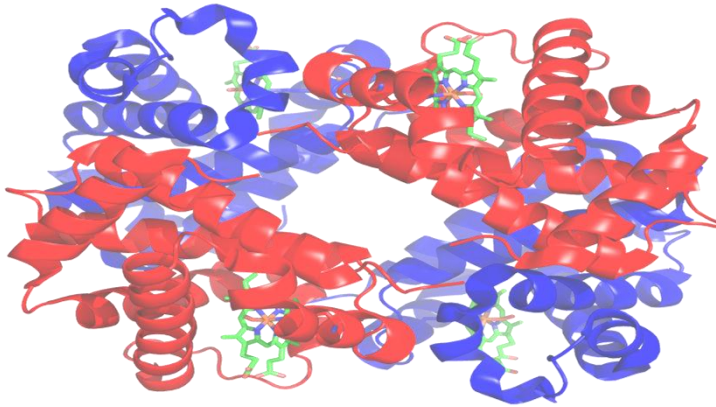


Figure 2.1

a) Discuss how the structure of haemoglobin is related to its function in a mammal. [3]

- Quaternary structure**
- It consists of four polypeptide chains: 2 alpha and 2 beta globin chains**
- each of has prosthetic haem group in the centre made up of porphyrin ring and iron**
- each iron binds to 1 molecule of O_2 => transport oxygen**
- Hydrophobic aa residues hidden on interior and hydrophilic aa residues on the exterior => soluble => able to dissolve in blood to transport oxygen.**

b) Hamoglobin is a globular protein. State 2 difference between fibrous and globular protein. [2]

Type	Nature	Function
Fibrous 	Secondary structure most important (little or no tertiary structure) Insoluble in water Physically tough Long parallel polypeptide chains cross-linked at intervals forming long fibres or sheets	Perform structural functions in cells and organisms, e.g. components of connective tissue, collagen (tendons, bone matrix), myosin (in muscle sarcomere), silk (spiders' webs), keratin (hair, horn, nails, feathers)
Globular 	Tertiary structure most important Polypeptide chains tightly folded to form spherical shape Easily soluble to form a colloidal suspension	Globulins of blood serum – important in immunology Form enzymes, antibodies and some hormones, e.g. insulin Important in protoplasm as they hold water and other substances and serve to maintain molecular organisation
Intermediate	Fibrous but soluble	e.g. fibrinogen – forms insoluble fibrin when blood clots

c) Predict what will happen if haemoglobin is exposed to high temperature and explain your answer. [3]

- **High temperature increases kinetic energy of protein molecules, resulting in violent vibration and**
- **breaking of bonds (hydrogen bonds, hydrophobic interactions, ionic bonds)
Disulphide bonds only break at very high temperature.**
- **Loss of 3D conformation of protein leads to loss of biological functions.**

[Total: 8 marks]

Question 3

A man and his wife have blood group AB. However, their first boy has blood group O. Upon consultation with a geneticist, they realised that this is due to the boy lacking the H antigen. This antigen is an essential precursor of the ABO blood group antigens. The boy is homozygous recessive at the H gene locus on chromosome 19. Hence, he cannot synthesise either the A or B antigen on his red blood cells even if he may have the I^A or I^B allele at the ABO gene locus on chromosome 9.

Using suitable symbols, draw a genetic diagram in the space provided below to show the expected blood group phenotypic ratio in the offspring of this couple. [5]

Parental Phenotype	Blood Grp AB	x	Blood Grp AB
Parental genotype	$HhI^A I^B$	x	$HhI^A I^B$
Gametes	HI^A HI^B hi^A hi^B		HI^A HI^B hi^A hi^B

Punnett Square

	HI^A	HI^B	hi^A	hi^B
HI^A	$HHI^A I^A$ Blood grp A	$HHI^A I^B$ Blood Grp AB	$Hhi^A I^A$ Blood grp A	$Hhi^A I^B$ Blood Grp AB
HI^B	$HHI^A I^B$ Blood Grp AB	$HHI^B I^B$ Blood Grp B	$Hhi^A I^B$ Blood Grp AB	$Hhi^B I^B$ Blood Grp B
hi^A	$Hhi^A I^A$ Blood Grp A	$Hhi^A I^B$ Blood Grp AB	$hhi^A I^A$ Blood grp O	$hhi^A I^B$ Blood Grp O
hi^B	$Hhi^A I^B$ Blood Grp AB	$Hhi^B I^B$ Blood Grp B	$hhi^A I^B$ Blood GrpO	$hhi^B I^B$ Blood Grp O

F1 phenotype: Blood grp O: A: B:AB
4: 3: 3:6

[Total: 5 marks]

Question 4

Rubisco is an important enzyme necessary for photosynthesis. One characteristic of this enzyme is its affinity with oxygen or carbon dioxide, leading to photorespiration and photosynthesis respectively. An investigation to determine the effect of oxygen to carbon dioxide ratio on the amount of carbohydrates produced was carried out. Table 4.1 tabulates the data collected.

Experiment No.	Light intensity / lux	Amount of CO ₂ available in stroma / %	Amount of O ₂ available in stroma / %	pH	Amount of carbohydrates produced / arbitrary units
1	20	10	0.002	3	0.01
2	20	10	0.002	8	250
3	20	2	0.009	3	0.01
4	20	2	0.009	8	15

Table 4.1

- a) With reference to Table 4.1, describe the effect of oxygen on the amount of carbohydrates at pH8. [1]
- **At pH8, as the amount of oxygen in stroma increases from 0.002% to 0.009%, the amount of carbohydrates produced decreases from 250au to 15 au.**
- b) Suggest a reason for the low amount of carbohydrates produced at pH3 even through the oxygen to carbon dioxide ratio is altered in experiment numbers 1 and 3. [2]
- **pH3 is beyond the optimum pH for the proper function of enzyme rubisco, where H⁺ concentration is high, H⁺ neutralises oppositely charged R groups involved in ionic bonding that holds the enzyme active site in specific conformation,**
 - **Hence substrate is no longer complementary to enzyme active site and thus cannot bind to enzyme to form carbohydrates.**

c) Explain the effect of high oxygen to carbon dioxide ratio on the amount of carbohydrates produced. [3]

- Oxygen competes with carbon dioxide for the active site of Rubisco / Oxygen is a competitive inhibitor of Rubisco.
- In the presence of high oxygen to carbon dioxide ratio, less carbon dioxide binds to active site of Rubisco,
- Leading to decreased carbon fixation, less glyceraldehyde-3-phosphate formed from glycerate-3-phosphate, thus less carbohydrates formed (15au as compared to 250au at pH 8).

d) Explain the importance of oxygen in photorespiration. [4]

- Oxygen serves as final electron acceptor at the end of the electron transport chain, accepting electrons and combining with H⁺ to form water via cytochrome oxidase.
- Electronegative oxygen helps to maintain the pull of electrons down the electron transport chain.
- As electrons are pulled down the ETC comprising electron carriers of progressively lower energy levels, energy released is released
- Energy released is used to maintain a proton gradient across inner mitochondrial membrane for ATP synthesis (chemiosmosis)

Question 5

When threatened, lizards lose their tails as a distraction to predators. The detached tail is able to move involuntarily due to muscle spasms for up to five minutes. The process of a lizard breaking off its tail is known as autotomy. The good news for the lizard is that it is able to regenerate its tail in less than a year as shown in Figure 5.1.

Interested to find out more about the regenerative ability of the lizard tail, scientists have isolated fast dividing cells in a regenerating lizard tail for more analysis.

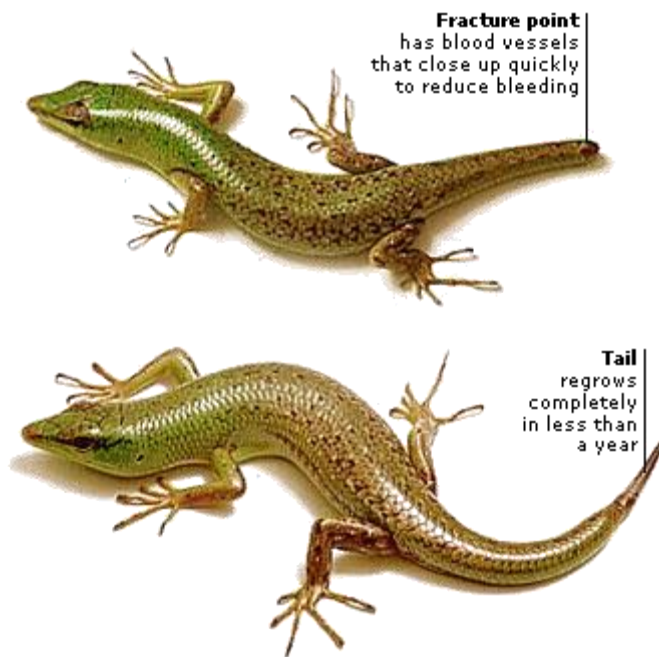


Figure 5.1

- a) State the cellular process that is responsible for the regeneration of the lizard tail. [1]
- **Mitosis**
- b) List four properties of the cells responsible in the regeneration of the lizard tail. [4]
- **Genetically identical (to parent cells/ each other);**
 - **Unspecialised/ Undifferentiated;**
 - **Capable of long-term self renewal;**
 - **Able to differentiate into specialised cells under appropriate conditions;**

c) Contrast the cellular process stated in **a)** to reductive nuclear division. [4]

Features	Mitosis	Meiosis
1. Occurrence	Occurs in _Somatic_ cells	Occurs in germline cells that produce gametes (in reproductive organs)
2. No. of nuclear division	1 nuclear division to form 2 daughter nuclei	2 successive nuclear divisions to form 4 daughter nuclei
3. Interphase and DNA replication	Interphase occurs once and DNA replication takes place once.	Interphase _may occur <i>once or twice</i> but DNA replication takes place only once, during the 1st interphase.
4. Prophase	- Homologous chromosomes do not pair up; Synapsis does not occur - No chiasmata formation and no crossing over occurs.	- Homologous Chromosomes pair up to form bivalents in prophase I; synapsis occurs - Chiasmata formation and crossing over between _non sister chromatids of homologous chromosomes take place; - This leads to new combinations of alleles.

- d) Most somatic cells in the human body, unlike the cells from the regenerating lizard tail, have a limited lifespan. However, certain changes to critical genes can occur over time to lead up to the formation of tumors in the body which can then become cancerous.
- i) Outline briefly how a tumor becomes cancerous. [2]
- Angiogenesis occurs/ blood vessels grow around tumor to supply nutrients to the growing tumor.
 - Tumor cells accumulate mutation that allows it to metastasize, entering the bloodstream to migrate to other parts of the body to form secondary tumors.

Essay (20 marks)

Question 6

a) Using a named example, describe how large quantities of a particular protein may be produced using bacteria systems. [10]

- **Isolate genomic DNA from liver or any part of the body*
- *Amplify gene for liver enzyme using PCR*
- *Design primers that flank either side of the gene*
- *Add dNTPs, Taq polymerase*
- *Allow for many cycles of denaturing, annealing and elongating in thermal cycler*
- *Cut amplified gene and plasmid with the same restriction enzyme*
- *Action of restriction enzyme produces complementary sticky ends, overhangs, H bonds, complementary base pairing*
- *Ligate gene and plasmid with action of ligase form covalent phosphodiester bond btwn adjacent nucleotides*
- *Use of expression vector*
- *Add ligation mix to bacteria and transform bacteria via heat shock method / electroporation*
- *Description of heat shock / electroporation*
- *Spread bacteria onto agar plates with ampicillin and nutrients / LB medium and allow to grow overnight*
- *Brief description of selection method – double selection, Beta-galactosidase activity or colony hybridisation: (1 mark)*
- *Culture selected bacteria in biofermenter*
- *Induce expression of liver gene*
- *Purify protein*

b) Describe the advantages and limitations of the Polymerase Chain Reaction (PCR). [4]

Advantage:

- **sensitive;**
- amplification possible from **trace amount of template DNA;**
- fast/efficient method of testing as 35 – 40 PCR cycles can be completed in several hours/(ref to short times required for each PCR segment); [R: “fast/efficient” without any elaboration];
- use of **thermostable** Taq polymerase allows for automation of procedure without need to add polymerase for every cycle;
- allows **amplification/replication** of selected target region of DNA flanked by/using specific PCR primers;
- **Programmable thermal cycler** allows automation of PCR;
- faster method of amplifying/cloning DNA sequence **than conventional cell-based cloning;**

Reject: create many copies of DNA using small sample, sensitive allowing amplification of small DNA

Disadvantage:

- sensitive to contamination of template DNA by **foreign/non-target nucleic acids;**
- may give false positives;
- requires careful handling to prevent contamination;
- by non-target nucleic acids;
- Use of **Taq DNA polymerase;** may result in misincorporation errors in PCR products; as (Taq DNA polymerase) lacks proof-reading activity / 3'→5' exonuclease activity (to correct misincorporated bases);
- Needs preliminary knowledge of part of/nature of target sequence; before specific primers can be designed;
- Mutations in regions complementary to primer(s) may cause reduction in **annealing/amplification efficiency;**
- PCR products labile/unstable as they are not associated with histones;

Accept any other logically elaborated points

Reject: primer sequence must be known, vague ref to small/many/large error rates/size limit of PCR products without ref to figures

c) Explain how the structure of starch and cellulose is linked to its function. [6]

Structure	Property	Function
Starch is a large molecule.	Insoluble and osmotically inactive	Ideal for <u>storage</u>
Composed of several hundred to several thousand glucose residues linked by $\alpha(1\rightarrow4)$ and $\alpha(1\rightarrow6)$ glycosidic bonds.	$\alpha(1\rightarrow4)$ glycosidic bonds are easily hydrolysed by amylase in plants and animals	Store large amounts of <u>energy</u> Glucose is the main respiratory substrate of organisms.
Chains of amylose are helical in shape.	<u>Helical</u> structure allows starch to be <u>compact</u>	Ideal for <u>storage</u>
Amylopectin is highly branched.	<u>Porous</u> this results in <u>large surface area</u> exposed to amylase for hydrolysis.	Increase rate of hydrolysis of amylopectin to yield energy
Hydroxyl groups of the glucose residues project into the interior of the helix.	<u>Intramolecular hydrogen bonds</u> stabilize helical structure Hydroxyl groups used to form hydrogen bonds are “shielded” in helix, making starch <u>insoluble</u>	Ideal for storage

Structure	Property	Function
Cellulose is a large molecule.	<u>Insoluble</u> , no effect on osmolarity of the cell	Maintain structure of plant cells
Composed of β -glucose units linked by $\beta(1\rightarrow4)$ glycosidic bonds.	Resistant to digestion (few organisms have specific enzymes to hydrolyse cellulose)	Maintain structure of plant cells
Chains of cellulose are long, straight and unbranched.	Able to form <u>long fibres</u>	Provide support to maintain structure of plant cells
Hydroxyl groups of the glucose residues project outward on both sides of the cellulose chains.	Facilitates formation of inter-chain hydrogen bonds between adjacent strands (crosslinking)	Provides high tensile strength so as to maintain structure of plant cells
Chains are arranged in groups to form microfibrils and macrofibrils.	Increase tensile and mechanical strength	Maintain structure of plant cells
Presence of <u>large intermolecular spaces</u> between macrofibrils	Completely permeable to water molecules	Allow movement of water and water soluble substances in and out of the cell while maintaining structure of the cell.

Question 7

Describe how the membrane affects the entry and exit of substances in and out of a cell.[6]

- Membrane is selectively permeable, due to the lipid bilayer and membrane proteins
- Hydrophobic / non-polar substances are able to dissolve in the lipid bilayer, and pass through the membrane down a concentration gradient
- Small molecules such as water and gases are able to pass through the membrane without dissolving
- Ions, hydrophilic/polar molecules and large molecules are unable to pass through the fatty acid tails of the bilayer
- Presence of trans-membrane proteins which selectively allow certain polar molecules and ions to pass through
- May occur down or against a concentration gradient (active transport)

Intrinsic proteins can act as ion channels, ligand-gated channels or protein pumps

- Ion channels will facilitated movement of substances across the membrane following laws of simple diffusion
- Ligand-gated channels will undergo conformational change to its structure in the presence of a stimulus, thus allowing movement of substances across the membrane
- While protein pumps, with the aid of ATP, transport ions/ substances across the membrane against concentration gradient

(b) Describe the main ways in which an enzyme differs from DNA. [5]

	Features	Enzyme	DNA
1	Overall structure	<u>Globular proteins</u> consists linear polypeptide chains folded into compact spherical shape	<u>Linear DNA</u> consists of a double polynucleotide chains wound into a <u>double helix</u> ;
2	Monomers	Proteins are made up of <u>amino acids</u>	DNA are made up of deoxyribonucleotides;
3	Types of different monomers.	There are <u>20 different common amino acids</u> present in proteins	There are <u>four</u> different nucleotides adenine, thymine, cytosine, guanine; ® uracil
4	Bonds btn monomers	Amino acids are held by <u>peptide bonds</u> between –COOH & -NH to form protein	Nucleotides are held by <u>phosphodiester linkages</u> between 3' –OH of 1 nucleotide and the phosphate group of another nucleotide;
5	Other bonds supporting overall structure	The globular structure of protein is maintained by <u>hydrogen bonds</u> , <u>ionic bonds</u> , <u>disulfide bonds</u> , and/or <u>hydrophobic interactions between R- groups</u>	The double helix structure of DNA is maintained by <u>hydrogen bonding between nitrogenous bases</u> and <u>hydrophobic interaction between stacked bases</u> ;
6	Properties of the molecules of the biomolecule	<u>Hydrophobic amino acids on the interior</u> and <u>hydrophilic amino acids on the exterior of protein</u> ;	<u>Hydrophobic nitrogenous bases on the interior</u> and <u>hydrophilic sugar-phosphate backbone on the exterior of protein</u> ;
9	Elements present	some enzymes contain element <u>sulphur</u> in some amino acids (e.g. cysteine)	DNA contains element <u>phosphorus</u> in phosphate group of the nucleotide ;

a) Relate the structure of DNA to its function. [7]

Function of DNA: To function as HEREDITARY MATERIAL [1/2 – MUST MENTION!]
[Reject: as genetic material]

1. In DNA replication

Structure:

- Double helix, [1/2]
- The sequence of bases on the two strands are complementary / Nitrogenous bases allow for highly specific base-pairing to occur between complementary bases. [1/2]
- Adenine pairs with Thymine & Guanine pairs with Cytosine [1/2]

Relate to function

→ each strand can act as a template to which a complementary set of nucleotides will attach by base pairing/ synthesize a new daughter strand [1/2]

→ Therefore each original DNA molecule can give rise to 2 copies which are identical in structure and base sequence /

→ allow for accurate replication to occur. [1/2]

2. In DNA repair

Structure:

- As the sequence of bases on the two strands are complementary / As nitrogenous bases allow for highly specific base-pairing to occur between complementary bases; [1/2; do not credit again if mention earlier]
- Adenine pairs with Thymine; Guanine pairs with Cytosine [1/2; do not credit again if mention earlier]

Relate to function

→ the intact complementary strand can be used as a template to guide DNA repair. [1/2]

→ Such repair mechanisms ensure that the integrity of the DNA sequence remains intact. [1/2]

[Reject: maintains integrity]

→ This is important as DNA functions as hereditary material. [1/2; do not credit again if mention earlier]

3. Stability of the DNA molecule

- Hydrogen bonds between complementary base pairs in the double helix/ complementary nitrogenous bases of adjacent DNA strands; [1/2]
- hydrophobic interactions between the stacked bases; [1/2]
- strong covalent phosphodiester bonds between adjacent nucleotides of the same DNA strand / within each strand; [1/2]
- deoxyribose sugar is less reactive than ribose sugar; [1/2]

- Purine always pairs with pyrimidine [1/2]
- Ensures width of DNA/ between the 2 sugar phosphate backbones is constant [1/2]
- stabilise the structure of the double helix [1/2] and
- maintains the integrity of the DNA sequence.
[1/2; do not credit if credited above.]
- This is important for DNA to function as a hereditary material.
[1/2; do not credit if credited above.]